

ANKARA ETIMESGUT, Turkey

WMO: 171290

Lat: 39.950N

Lon: 32.689E

Elev: 809

StdP: 91.98

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.9	-7.7	-13.2	1.3	-6.7	-11.2	1.6	-4.6	9.4	7.7	8.1	6.6	0.8	280	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	35.0	17.6	33.2	17.5	31.8	17.2	20.0	29.8	19.1	28.8	18.4	28.2	3.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	13.2	23.4	15.8	12.4	23.0	14.9	11.7	22.5	60.6	30.0	57.7	28.8	55.2	28.0	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	6.9	6.0	-12.8	37.4	3.0	1.6	-14.9	38.5	-16.6	39.5	-18.3	40.4	-20.4	41.6		
(5)			-13.7	21.1	3.2	1.5	-16.0	22.2	-17.9	23.1	-19.7	23.9	-22.0	25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	1.2	2.7	6.4	11.0	16.2	20.2	23.7	23.9	18.9	13.3	6.7	2.7		
	DBStd	8.74	4.33	4.20	3.97	3.62	3.23	2.80	2.56	2.55	3.27	3.74	3.75	3.96		
	HDD10.0	981	274	204	122	29	1	0	0	0	14	109	228			
	HDD18.3	2679	532	437	369	221	82	13	1	32	158	348	485			
	CDD10.0	1826	1	1	11	59	192	306	426	432	268	117	11	1		
	CDD18.3	481	0	0	0	1	16	69	168	175	50	3	0	0		
	CDH23.3	6089	0	0	2	29	272	862	2058	2057	714	96	0	0		
	CDH26.7	2554	0	0	0	2	61	301	962	961	251	17	0	0		
(14) Wind	WSAvg	2.1	1.8	2.0	2.4	2.3	2.3	2.5	2.8	2.6	2.1	1.7	1.3	1.6		
(15) Precipitation	PrecAvg	375	44	34	34	42	47	33	14	12	17	24	29	48		
	PrecMax	626	99	90	83	94	89	92	51	43	95	69	117	97		
	PrecMin	248	4	7	8	6	17	2	0	1	0	0	2	5		
	PrecStd	78	28	23	20	25	19	27	15	11	20	19	23	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	17.1	22.0	26.6	30.8	34.0	37.7	37.0	34.1	28.9	21.0	14.2		
		MCWB	7.9	9.3	10.1	13.7	16.3	17.7	17.7	17.7	16.8	14.6	11.6	9.4		
	2%	DB	10.8	14.0	18.8	23.8	28.2	31.9	35.2	35.0	31.7	25.9	18.2	11.9		
		MCWB	7.1	8.0	9.4	12.7	15.9	17.2	18.0	17.4	16.2	13.6	10.7	8.2		
	5%	DB	9.0	11.9	16.3	21.2	26.4	30.0	33.2	33.2	29.8	23.8	16.0	10.1		
		MCWB	6.0	6.9	8.5	11.7	15.2	16.7	17.7	17.5	15.7	13.4	9.5	7.3		
	10%	DB	7.1	9.8	14.2	19.1	24.2	28.2	31.8	31.8	27.7	21.2	13.9	8.8		
		MCWB	5.1	6.0	7.8	10.8	14.1	16.1	17.5	17.3	15.1	12.6	8.8	6.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.6	10.5	11.5	15.2	18.9	20.0	22.2	20.9	19.4	16.1	13.0	10.2		
		MCDB	12.1	15.2	19.5	22.7	27.8	29.7	32.2	31.0	29.1	22.9	18.7	13.3		
	2%	WB	7.3	8.5	10.1	13.5	16.9	18.7	20.2	19.9	17.9	14.9	11.5	8.7		
		MCDB	10.2	12.8	16.7	21.8	25.9	28.1	30.1	29.7	27.4	21.9	16.6	11.2		
	5%	WB	6.2	7.3	9.1	12.3	15.8	17.7	19.1	19.0	16.8	14.2	10.4	7.7		
		MCDB	8.3	11.2	15.1	19.8	24.0	26.7	29.0	28.6	26.2	21.5	15.0	9.9		
	10%	WB	5.1	6.1	8.3	11.3	14.9	17.0	18.3	18.4	15.9	13.4	9.2	6.6		
		MCDB	6.9	9.3	13.6	18.1	22.6	25.6	28.4	28.1	25.2	19.9	13.1	8.5		
(35) Mean Daily Temperature Range	MDBR		7.8	10.1	11.7	12.6	13.4	14.0	14.7	14.5	15.0	13.5	12.1	8.0		
	5% DB	MCDBR	10.7	14.0	16.1	16.8	16.7	16.9	16.9	16.7	18.4	17.5	16.0	10.3		
		MCWBR	7.3	8.6	8.6	8.0	6.9	6.0	5.5	5.4	7.0	7.9	8.9	6.8		
	5% WB	MCDBR	9.4	12.2	13.8	14.9	14.8	14.7	14.8	14.4	15.3	14.2	13.5	9.1		
		MCWBR	6.7	7.7	7.6	7.6	6.5	5.8	5.6	5.2	6.4	6.8	7.9	6.2		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.333	0.376	0.413	0.392	0.372	0.370	0.368	0.348	0.348	0.318	0.307		
	taud		2.380	2.326	2.223	2.148	2.274	2.349	2.354	2.337	2.390	2.387	2.446	2.436		
	Ebn at Noon		854	879	872	861	886	902	901	894	893	850	838	827		
	Edh at Noon		85	103	128	147	133	123	122	120	107	96	79	75		
(44) All-Sky Solar Radiation	RadAvg		1.88	2.88	4.03	5.17	6.14	6.98	7.40	6.59	5.19	3.55	2.45	1.67		
	RadStd		0.23	0.41	0.36	0.46	0.44	0.58	0.31	0.27	0.35	0.25	0.24	0.21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page